ECOLOOP — 8-MINUTE SOLUTION PITCH SCRIPT

HACK4SDG: Ideas Today, Sustainable Tomorrow • Nitte University, Bengaluru Campus

Team Name:	Power Rangers (Entry #077)	**Focus SDG:**	Goal 12: Responsible Consumption & Production
Pitch Format:	4 Speakers • 8 Minutes Total (~2 min/speaker)	**Live Platform:**	**ecoloop-campus.vercel.app**

100-POINT EVALUATION CRITERIA MATRIX (5 Sections × 20 Points)

- Speaker 1 [SDG Relevance & Problem]: Hostel Drawer Reality, First-Mile Gap, Informal Sector Toxicity (20 Pts)
- Speaker 2 [Innovation & Originality]: Tamper-Evident 5-Stage Hardware Passport, Mobile Web MVP (20 Pts)
- Speaker 3 [Social & Environmental Impact]: GEM 2024 Scope-3 (1.44 kg CO2e/kg), NAAC 7.1.3 Compliance, DPDP Act (20 Pts)
- Speaker 4 [Feasibility & Scalability]: Zero-Hardware Pilot, Scrap Monetization, Inter-College Network (40 Pts)

SPEAKER 1: THE GROUNDED STUDENT REALITY & HIDDEN ROOM CRISIS

0:00 – 2:00 | SLIDE 01

"Respected judges, faculty coordinators, and fellow innovators from Nitte University. Good morning. We are team Power Rangers, presenting EcoLoop.

Let me start not with global statistics, but with something every single student and warden in our hostels knows too well: the bottom drawer of our study desks.

Walk into Nandani hostel, Kaveri, or Yourspace right now. In nearly every room, you will find a swollen laptop battery, frayed fast-charging cables, burnt out earplugs, or dead power banks. When we surveyed our fellow hostel mates, we asked a simple question: "Why haven't you discarded that broken phone or dead keyboard?"

Nobody said, "I want to pollute the earth." They gave us three deeply honest reasons:

1. Convenience: "Where do I even take this on campus? There's no designated e-waste bin near my wing."
2. Data & Trust Hesitation: "My old phone has my personal photos and passwords. If I give it to a casual scrap collector at the campus gate, where does it end up?"
3. The Inertia of Occasional Drives: Colleges organize a green drive once a year. If my charger dies in February, it sits in my drawer until November—or worse, gets tossed into the hostel mixed waste dustbin with wet food waste.

This is the root cause. India does not lack certified CPCB recyclers. What we lack is the first-mile bridge between a student's room and authorized dismantlers.

When students discard e-waste into common bins, it falls into the informal sector. In peri-urban Bengaluru, informal handlers acid-leach circuit boards and burn PVC cables in open fires to extract copper, releasing lead, mercury, and dioxins into our groundwater. This directly violates UN SDG 12: Responsible Consumption and Production. EcoLoop closes this exact first-mile loophole—permanently."

SPEAKER 2: THE SOLUTION — DIGITAL PASSPORT & WORKING PROTOTYPE 2:00 – 4:00 | SLIDES 02 & 03

"Thank you, Speaker 1. To solve this, we didn't just design a presentation—we built and deployed a fully working web platform called EcoLoop, currently live at ecoloop-campus.vercel.app.

EcoLoop replaces vague promises with a tamper-evident 5-stage Hardware Passport:

[Registered '" Scheduled '" Collected '" Handed Over '" Verified Recycled]

Here is how simple and grounded it is in a student's daily routine:

1. Self-Service Logging in 30 Seconds: A student opens the web app on their phone. Using our Master Catalog of over 55 common campus devices, they select their device—say, a defective laptop. The system instantly displays its calibrated weight and standard recoverable fractions according to CPCB guidelines.
2. Unique QR Hardware Passport: The platform generates a unique, scannable QR token for that specific item. No confusion, no unlabelled mystery boxes.
3. Calibrated Drop-off or Doorstep Pickup: The student either drops it at their hostel reception box or schedules an EcoSquad collector slot.
4. Instant Weight Verification: The campus collection team scans the QR using any phone camera via our integrated scanner, validates the actual weight, and the student instantly receives verified reward points.
5. Dismantling Certificate: When the batch is handed over to our authorized CPCB recycler, the final green recycling certificate is digitally issued back to the student's dashboard.

How does this differ from existing alternatives?

- Traditional e-waste drop-boxes are passive black holes. You drop a phone in; you have no idea if it was stolen by housekeeping or actually recycled.
- One-time annual drives lack habit continuity.
- Government EPR portals operate exclusively at industrial B2B scale.

EcoLoop complements the CPCB EPR framework at the micro-campus level. It brings transparency, dignity, and real-time feedback to everyday civic recycling."

SPEAKER 3: ENVIRONMENTAL EVIDENCE, PRIVACY & INSTITUTIONAL VALUE 4:00 – 6:00 | SLIDES 03 & 04

"Thank you. Now, let's look at the numbers and the social reality. Why does this matter to Nitte University and our broader urban community?

First: Empirical Environmental Impact.

Every kilogram of electronics diverted from a landfill directly prevents hazardous contamination: Circuit boards contain lead solder, beryllium, and cadmium that bioaccumulate in soil.

Using the GEM 2024 Scope-3 emission factor of 1.44 kg CO_2e saved per kg of e-waste recycled, diverting just 500 kilograms of campus e-waste prevents over 720 kilograms of greenhouse gas emissions—the equivalent of planting 32 mature trees or offsetting 4,000 km of car travel. More importantly, it reclaims high-density secondary raw materials: engineering plastics, copper wiring, and precious circuit metals.

Second: Student Privacy by Design.

Under India's Digital Personal Data Protection (DPDP) Act 2023, students are rightfully worried about personal data. EcoLoop embeds mandatory certified media-sanitization checklists before item dispatch, ensuring peace of mind. Furthermore, when campus telemetry is shared with municipal bodies or research labs, all personal student identifiers are cryptographically stripped.

Third: Direct Institutional Value for Nitte University.

This is not an isolated student experiment. Colleges in India face rigorous NAAC Accreditation criteria. NAAC Criterion 7.1.3 specifically demands institutional proof of "Facilities for alternate sources of energy and waste management, including green campus initiatives and e-waste management."

Today, universities struggle to present verifiable documentation during audit inspections. EcoLoop features an automated One-Click NAAC Criterion 7 Audit Dossier generator that compiles audited manifests, collection logs, and authorised recycler receipts into an inspection-ready report for the administration."

SPEAKER 4: GROUNDED ECONOMICS, FEASIBILITY & THE NATIONAL VISION 6:00 – 8:00 | SLIDES 04 & 05

"Thank you, Speaker 3. A solution can be noble, but if it is not economically viable and operationally feasible, it will fail after hackathon day. Here is our grounded implementation model:

Phase 1: Zero-Hardware MVP Pilot (Weeks 1–4)

We do not ask for expensive IoT bins on Day 1. EcoLoop is already deployed as a lightweight progressive web app on Vercel with a resilient zero-config backend. We start with two designated pilot blocks on our campus: one residential block—say, Nandani / Yourspace—and one academic laboratory cluster. Existing department drop-points are designated with QR manifest boards, managed by student EcoSquad volunteers.

Phase 2: Self-Sustaining Financial Model

How does EcoLoop sustain itself without burdening university funds?

- Certified CPCB dismantlers pay bulk salvage rates for verified, segregated e-waste lots (averaging Rs 25 to Rs 45 per kg depending on ferrous and PCB grade).
- A fraction of this liquidation revenue funds student eco-rewards (redeemable at campus cafeterias, stationery stores, or library printing quotas).
- The remaining revenue maintains platform cloud costs and supports university green club drives.

Phase 3: Scalability Beyond Our Campus (Weeks 5–12)

Once stabilized at Nitte Bengaluru, the multi-tenant architecture of EcoLoop scales horizontally:

- Inter-College Green Leagues: Neighboring institutions like MIT Manipal, NITK, or BMS can participate in automated inter-collegiate diversion leaderboards.
- Municipal Hotspot Synergy: Our newly integrated Municipal Portal module allows municipal corporations (like BBMP and MCC) to request coordinated bulk community drives in documented e-waste hotspots.

To conclude: The challenge of e-waste is not waiting for a breakthrough in physics. It is waiting for human-centric software that makes doing the right thing as easy as ordering food on your phone.

EcoLoop turns student rooms from forgotten storage dumps into the starting point of India's circular economy. We have the working code, we have the verified pilot blueprint, and we are ready to take Nitte University to the national finals. Thank you, and we welcome your questions!"

JUDGES Q&A ANTICIPATED DEFENSE STRATEGY (CRITICAL HACKATHON RUBRICS)

Q1: What prevents students from dropping common trash or cheating weight?

A1: Weight validation happens in-person during calibrated scale handover with the EcoSquad volunteer. The QR code links to the pre-logged item category, and points are only accredited once the calibrated scale confirms the physical item.

Q2: How do you handle swollen lithium-ion battery hazards and fire safety?

A2: The Master Recovery Catalog automatically flags hazardous and swollen battery categories, issuing immediate user isolation instructions and routing items to dedicated vermiculite-insulated collection bins.

Q3: Why would university administration adopt EcoLoop instead of current tenders?

A3: Universities spend hundreds of hours manually preparing NAAC Criterion 7.1.3 e-waste compliance audits. EcoLoop gives the administration an automated, tamper-evident audit ledger with zero extra software spend.